Biomedical Entity Embeddings via Contrastive UMLS Alignment

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GitHub Repository
Project Files

Abstract

We present a systematic study of biomedical text embeddings trained from scratch and progressively enriched with knowledge from the Unified Medical Language System (UMLS). Both a Word2Vec encoder and a custom Transformer encoder are pre-trained on PubMed abstracts and subsequently fine-tuned through contrastive alignment using three sources of UMLS structure: synonym pairs, semantic type supervision, and relational triples. The resulting embeddings are evaluated on four downstream clinical NLP tasks—natural language inference (NLI4CT), disease entity linking (NCBI Disease, BC5CDR-disease), and semantic textual similarity (BIOSSES)—as well as on three internal UMLS probing benchmarks covering link prediction, relational retrieval, and semantic type classification. We show that synonym alignment is a necessary but insufficient condition for strong performance, while relation-based contrastive learning provides the largest single gain on retrieval tasks ($\times$10.5 MRR for the Transformer). A classification–retrieval trade-off is identified and shown to be architecture-agnostic. SapBERT, a domain-expert BERT pre-trained with UMLS synonyms, serves as a competitive reference baseline throughout.

1 Introduction

Biomedical natural language processing requires embeddings that capture the rich semantic structure of clinical and scientific language: that *myocardial infarction* and *heart attack* refer to the same concept, that *aspirin* can *treat headache*, and that *BRCA1* is a *Gene or Genome*. General-purpose embeddings trained on broad corpora (Wikipedia, Common Crawl) are known to underperform on biomedical tasks, motivating domain-specific pre-training on corpora such as PubMed pubmedbert.

A complementary line of work uses the UMLS knowledge base—a curated multi-vocabulary thesaurus of medical concepts—to provide structured supervision during embedding training. SapBERT sapbert is the canonical example: it fine-tunes PubMedBERT with NT-Xent contrastive loss on UMLS synonym pairs and achieves state-of-the-art entity linking on several benchmarks.

In contrast to these large pre-trained models, our work trains two smaller architectures—Word2Vec and a lightweight Transformer—entirely from scratch on PubMed text, then applies the same structured UMLS alignment pipeline to both. This allows a direct, controlled study of:

1. how much of the embedding quality is attributable to architecture vs. the alignment strategy;
2. which UMLS signal (synonyms, types, or relations) contributes most to each downstream task.

SapBERT is included for reference throughout to contextualise how far our lightweight trained-from-scratch models fall from a large pre-trained baseline.

2 Datasets

2.1 Pre-training Corpus: PubMed

Both the Word2Vec and Transformer baselines are trained on PubMed abstracts. The corpus is preprocessed with a simple tokeniser: text is lowercased and tokens shorter than two characters are discarded. Vocabulary construction uses a minimum frequency threshold of 2 and a maximum vocabulary size of 250,000 (Word2Vec) or 50,000 (Transformer). The Word2Vec training uses the full tokenised corpus; the Transformer is trained on a randomly sampled subset of 200,000 sentences to keep training tractable.

2.2 Alignment Data: UMLS

The Unified Medical Language System (UMLS) organises biomedical concepts into Concept Unique Identifiers (CUIs), each of which may have many surface forms (synonyms) across different medical vocabularies. We use three UMLS relational files:

`MRCONSO.RRF`
: Concept names and sources. Used to build synonym pairs: any two English-language surface forms sharing the same CUI form a positive pair for contrastive alignment. Up to 10 pairs per CUI are retained. Pairs are filtered so that at least one token appears in the model's vocabulary.

`MRSTY.RRF`

Semantic type assignments. Each CUI carries one or more semantic type labels (e.g., *Disease or Syndrome*, *Gene or Genome*). We retain the top 25 most frequent types (covering 486,528 concept–term pairs) and use them as multi-label targets for the auxiliary type classification head.

`MRREL.RRF`

Relations between concepts. We extract three clinically salient relation types: `has_manifestation`, `may_treat`, and `finding_site_of`. These form relation-based positive pairs used in the enhanced contrastive objective. The resulting dataset covers 282,126 relation pairs (141,063 positive + 141,063 random-negative) and 52,280 unique test anchor queries.

2.3 Downstream Task Datasets

NCBI Disease. The NCBI Disease corpus ncbi consists of 793 PubMed abstracts manually annotated with disease mentions, each normalised to an OMIM or MeSH identifier. We evaluate on the standard test split (100 documents, 960 mention–concept pairs) and search over the full CTD disease vocabulary of ≈13,000 entries. Because we use the full KB rather than a corpus-specific candidate list, our absolute scores are lower than published figures—this is a known and expected discrepancy arising from the harder retrieval setting.

BC5CDR-Disease. The BioCreative V Chemical–Disease Relation (BC5CDR) dataset bc5cdr contains 1,500 PubMed articles annotated with chemical and disease mentions normalised to MeSH. We evaluate the disease sub-task (BC5CDR-d) only, using the same full CTD KB and cosine-nearest-neighbour pipeline as NCBI Disease. We do not report results for the chemical sub-task: chemical names fall almost entirely outside the vocabularies of both our PubMed-trained Word2Vec (which covers general biomedical text rather than chemical nomenclature) and our compact 50,000-token Transformer, producing uniformly near-zero scores across all models. We acknowledge this as a limitation of our vocabulary design.

BIOSSES. The Biomedical Semantic Similarity Estimation System dataset biosses contains 100 sentence pairs drawn from MEDLINE citations, each rated by five domain experts on a 0–4 similarity scale. The task is to predict the mean human rating from the cosine similarity between sentence embeddings. We report Pearson r and Spearman ρ on the full set.

NLI4CT. NLI4CT nli4ct is a three-class natural language inference dataset (Entailment / Neutral / Contradiction) constructed from clinical trial reports. Each instance pairs a premise drawn from a clinical trial section with a hypothesis; the model must classify their logical relation. As only the training split is publicly available, we use a stratified 80/20 internal split (seed 42) throughout, yielding ≈1,600 training and 400 test examples.

3 Model Architectures

3.1 Word2Vec Encoder

The Word2Vec model uses the skip-gram architecture with negative sampling (SGNS) word2vec. The core idea is to train a word embedding matrix by predicting the surrounding context of each word: given a centre word w_I, the model should assign high probability to actual context words w_O and low probability to randomly drawn noise words. Two weight matrices are learned: an input (centre-word) matrix and an output (context-word) matrix, both of dimension $|\mathcal{V}| \times 300$.

Formally, for each observed (w_I, w_O) co-occurrence within a window of size 5, the SGNS objective maximises:

$$\mathcal{L}_{\text{SG}} = \log \sigma\big(\mathbf{v}_{w_O}^{\top} \mathbf{v}_{w_I}\big) + \sum_{k=1}^{K} \mathbb{E}_{w_k \sim P_n} \big[\log \sigma\big(-\mathbf{v}_{w_k}^{\top} \mathbf{v}_{w_I}\big)\big] \,, \tag{1}$$

where σ is the sigmoid function, K=10 is the number of noise samples, and $P_n(w) \propto f(w)^{3/4}$ is the smoothed unigram distribution. After training for 5 epochs across the full PubMed corpus, only the input matrix is retained as the embedding lookup. The vocabulary covers 250,000 tokens with a minimum word count of 5.

Inference. A text sequence is encoded by looking up each token's embedding and computing the arithmetic mean of all in-vocabulary tokens:

$$\mathbf{h}_{\text{W2V}}(\text{text}) = \frac{1}{|\mathcal{T}|} \sum_{w \in \mathcal{T}} \mathbf{v}_w \;\in\; \mathbb{R}^{300}, \tag{2}$$

where $\mathcal{T}$ is the set of in-vocabulary tokens in the input. Out-of-vocabulary tokens are silently ignored; if all tokens are OOV the representation is the zero vector. Before contrastive alignment the resulting vectors are L2-normalised to the unit sphere to prevent scale differences from dominating the cosine similarity computation.

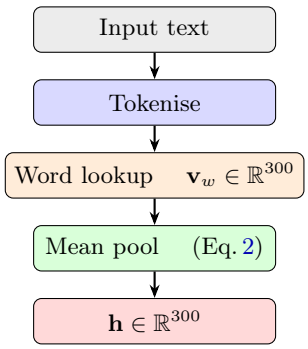

Figure 1: Word2Vec inference pipeline. Vectors are L2-normalised before contrastive alignment (Section 4).

3.2 Custom Transformer Encoder

The Transformer is a compact bidirectional encoder trained from scratch on PubMed text. We deliberately keep it small—2 layers, 4 attention heads, hidden size 128, feed-forward dimension 256—so that training and inference remain tractable without a GPU cluster. Despite its size it serves as a controlled testbed for studying whether contextual representations benefit from UMLS alignment differently than static Word2Vec embeddings.

Pre-training objective. The model is pre-trained with masked language modelling (MLM). Each input token is independently masked with probability $p_m = 0.15$ and replaced by the special `[MASK]` token. Let $\mathcal{M} \subseteq \{1, \ldots, T\}$ be the set of masked positions. The model must predict the original token at each masked position from the contextualised hidden states:

$$\mathcal{L}_{\text{MLM}} = -\frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \log \frac{\exp\big(\mathbf{W}_{\text{LM}} \mathbf{h}_t\big)_{x_t}}{\sum_{v=1}^{|\mathcal{V}|} \exp\big(\mathbf{W}_{\text{LM}} \mathbf{h}_t\big)_v}, \quad (3)$$

where $\mathbf{h}_t \in \mathbb{R}^{128}$ is the encoder output at position t and $\mathbf{W}_{\text{LM}} \in \mathbb{R}^{|\mathcal{V}| \times 128}$ is the language model head. Training runs for 1 epoch over 200,000 PubMed sentences with a batch size of 256 and learning rate 5×10^{-4} (AdamW).

Input representation. Each input sentence $(x_1, \ldots, x_T)$ is prepended with a `[CLS]` token, giving a sequence of length $T{+}1$. Token and position embeddings are summed element-wise:

$$\mathbf{e}_t = \mathbf{E}_{\text{tok}}[x_t] + \mathbf{E}_{\text{pos}}[t], \quad t = 0, 1, \ldots, T, \quad (4)$$

where $\mathbf{E}_{\text{tok}} \in \mathbb{R}^{|\mathcal{V}| \times 128}$, $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{32 \times 128}$, and the maximum sequence length is 32 tokens (longer sequences are truncated).

Encoder layers. The summed embeddings pass through $L = 2$ identical Transformer encoder layers. Each layer applies the following operations in sequence:

1. **Multi-head self-attention** (4 heads, head dimension 32):

   $$\text{MHA}(\mathbf{E}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_4)\, \mathbf{W}^O,$$

   where each head computes scaled dot-product attention over the full sequence. Padding positions are masked out via a key-padding mask.
2. **Residual connection + Layer normalisation.**
3. **Position-wise feed-forward network** ($128 \to 256 \to 128$) with GELU activation: $\text{FFN}(\mathbf{h}) = \mathbf{W}_2\, \text{GELU}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2$.
4. **Residual connection + Layer normalisation.**

Pooling. The sentence representation is the hidden state of the `[CLS]` token at the output of the final encoder layer: $\mathbf{h}_{\text{CLS}} \in \mathbb{R}^{128}$. The pre-trained baseline outputs $\mathbf{h}_{\text{CLS}}$ directly for downstream use. Aligned variants route $\mathbf{h}_{\text{CLS}}$ through the projection head (Section 4) before computing similarity scores.

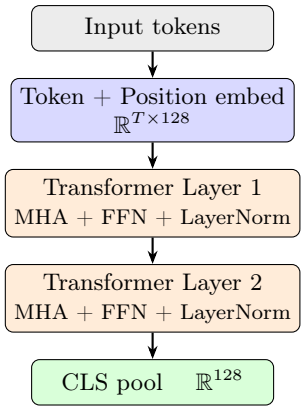

Figure 2: Custom Transformer encoder (2 layers, 128 hidden, 4 heads, GELU). Residual connections and layer normalisation are applied inside each block.

4 Contrastive Alignment Framework

Both pre-trained encoders produce a fixed-size vector—$\mathbf{h} \in \mathbb{R}^{300}$ for Word2Vec and $\mathbf{h} \in \mathbb{R}^{128}$ for the Transformer—but the metric structure of these spaces is not optimised for medical concept similarity. We apply a contrastive alignment stage that maps each encoder's output through a shared *projection head* and then trains on three complementary UMLS signals.

4.1 Projection Head

Motivated by SimCLR simclr, we attach a two-layer non-linear projection head on top of each backbone. The projection head serves two purposes. First, it decouples the representation used for contrastive training from the backbone's own feature space: the backbone is free to learn general-purpose features while the projection head absorbs alignment-specific geometry. Second, the non-linear bottleneck prevents *dimensional collapse*—a pathology where all representations converge to a low-rank manifold.

Formally, given a backbone representation $\mathbf{h} \in \mathbb{R}^d$, the head computes

$$\mathbf{g}(\mathbf{h}) = \mathbf{W}_2\, \text{ReLU}\big(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1\big) + \mathbf{b}_2, \quad (5)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d \times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{256 \times d}$. and the final projection is L2-normalised to the unit hypersphere:

$$\mathbf{z} = \frac{\mathbf{g}(\mathbf{h})}{\|\mathbf{g}(\mathbf{h})\|_2} \in \mathbb{R}^{256}. \quad (6)$$

All alignment objectives are applied directly to $\mathbf{z}$; the backbone representations $\mathbf{h}$ are *not* used by any loss term.

For Word2Vec the backbone is **frozen** during alignment (d=300): only $\mathbf{W}_1$, $\mathbf{b}_1$, $\mathbf{W}_2$, $\mathbf{b}_2$ are updated. For the Transformer the entire encoder is jointly fine-tuned alongside the projection head (d=128).

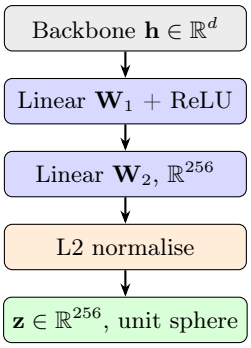

Figure 3: Projection head appended to both encoders during alignment. The L2-normalised output $\mathbf{z}$ is the representation used by all contrastive objectives.

4.2 Self-Supervised Alignment with UMLS

A central contribution of this work is framing UMLS knowledge as a source of *self-supervised* supervision signals. No human annotations are needed: positive pairs are derived automatically from the curated structure of UMLS.

Synonym pairs (MRCONSO). Each UMLS concept has multiple surface forms. Any two English-language strings sharing the same CUI constitute a positive pair; they should map to nearby points on the unit sphere. Up to 10 pairs are sampled per CUI to balance coverage with over-representation of prolific concepts. Pairs are filtered to ensure at least one token is in-vocabulary.

Relation pairs (MRREL). Concepts linked by a clinically salient relation—`has_manifestation`, `may_treat`, `finding_site_of`—form a second class of positive pairs. Their names are treated as anchor–target text pairs whose projections should also be pulled together, but through a *separate* contrastive loss term so that relational proximity does not overwrite synonym proximity.

In both cases the negatives are *in-batch*: all other samples in the mini-batch serve as implicit negatives, exactly as in NT-Xent. This removes the need for explicit negative mining.

4.3 NT-Xent Contrastive Loss

Let (x_i^a, x_i^b) be the i-th positive pair in a mini-batch of N pairs. After projection we have $2N$ unit vectors $\{\mathbf{z}_1^a, \mathbf{z}_1^b, \ldots, \mathbf{z}_N^a, \mathbf{z}_N^b\}$. For a query vector $\mathbf{z}_i^a$, its positive is $\mathbf{z}_i^b$ and the remaining $2N-2$ vectors act as negatives. The per-sample NT-Xent loss is

$$\ell(i, a \to b) = -\log \frac{\exp\big(\mathbf{z}_i^a \cdot \mathbf{z}_i^b / \tau\big)}{\sum_{\substack{j=1 \\ j \neq i}}^{N} \Big[\exp\big(\mathbf{z}_i^a \cdot \mathbf{z}_j^a / \tau\big) + \exp\big(\mathbf{z}_i^a \cdot \mathbf{z}_j^b / \tau\big)\Big]}, \tag{7}$$

where τ=0.07 is the temperature. The full symmetric loss averages over both directions:

$$\mathcal{L}_{\text{NT-Xent}} = -\frac{1}{2N} \sum_{i=1}^{N} \Big[\, \ell(i, a \to b) + \ell(i, b \to a) \,\Big]. \tag{8}$$

This objective is applied independently to synonym pairs ($\mathcal{L}_{\text{NT-Xent}}^{\text{syn}}$, batch size 512 for Word2Vec, 256 for Transformer) and to relation pairs ($\mathcal{L}_{\text{NT-Xent}}^{\text{rel}}$, 30 % sampling ratio per batch).

4.4 Semantic Type Classification Loss

Beyond pairwise proximity, UMLS assigns each concept to one or more of 25 semantic types (e.g., *Disease or Syndrome*, *Pharmacologic Substance*). We attach a linear *type classifier* $f_{\text{type}} : \mathbb{R}^{256} \to \mathbb{R}^{25}$ directly on the projection $\mathbf{z}$ and train it with weighted multi-label binary cross-entropy:

$$\mathcal{L}_{\text{type}} = -\frac{1}{25} \sum_{c=1}^{25} w_c \Big[y_c \log \sigma(\hat{y}_c) + (1 - y_c) \log\big(1 - \sigma(\hat{y}_c)\big) \Big], \tag{9}$$

where $y_c \in \{0, 1\}$ is the ground-truth type indicator and $\hat{y}_c = f_{\text{type}}(\mathbf{z})_c$. Class weights $w_c \propto 1/f_c^{0.5}$ (inverse square-root frequency) compensate for type imbalance; the weights are computed once from the training split and held fixed throughout.

4.5 Combined Objective and Warmup Schedule

The full training objective sums all active loss terms:

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{NT-Xent}}^{\text{syn}} + \mathcal{L}_{\text{NT-Xent}}^{\text{rel}} + \lambda(t) \cdot \mathcal{L}_{\text{type}}} \tag{10}$$

The relation term and type term are activated selectively depending on the ablation variant (Table 1). Because the type classification signal can dominate early in training—when synonym geometry is not yet stable—we apply a linear warmup schedule to the type weight:

$$\lambda(t) = \lambda_0 \cdot \min\left(1, \frac{t}{T_{\text{warm}}}\right), \quad \lambda_0 = 0.1, \tag{11}$$

where t is the current training step and T_{warm} is 20 % of the total training steps. The synonym contrastive term is always active whenever alignment training is performed.

Table 1: Active loss terms per ablation variant. ✓= active.

Variant	$\mathcal{L}^{\text{syn}}$	$\mathcal{L}^{\text{rel}}$	$\lambda\mathcal{L}_{\text{type}}$
Baseline			
UMLS	✓		
Synonym Only	✓		
Syn + Type	✓		✓
Syn + Relation	✓	✓	
Full Enhanced	✓	✓	✓

The difference between the *UMLS* and *Synonym Only* variants is one of data quality: *Synonym Only* applies vocabulary-based filtering to retain only pairs where both strings contain at least one in-vocabulary token, whereas *UMLS* uses all synonym pairs verbatim. This tests whether cleaner alignment pairs improve downstream task scores.

5 Downstream Task Evaluation

All models are evaluated in a **frozen-embedding** regime: backbone weights are locked after training and the resulting vectors are used directly for retrieval (cosine nearest-neighbour) or fed into a lightweight probe (MLP classifier for NLI). This isolates embedding quality from task-specific fine-tuning and gives a clean geometric audit of each vector space. SapBERT is included as an upper-bound reference throughout.

5.1 NCBI Disease Entity Linking

Given a raw disease mention, the model must retrieve its canonical MeSH/OMIM identifier from the full CTD vocabulary (≈13,000 entries) using cosine nearest-neighbour search. Table 2 reports Acc@1, Acc@5, and MRR on the standard 100-document test split.

Table 2: NCBI Disease entity linking. Best trained-from-scratch in bold.

Model	Acc@1	Acc@5	MRR
SapBERT	0.607	0.730	0.664
W2V Baseline	0.284	0.416	0.341
W2V + UMLS (Syn)	**0.313**	**0.452**	**0.374**
W2V + Full Enhanced	0.298	0.444	0.358
Transformer Baseline	0.148	0.160	0.162
Transformer + UMLS (Fast)	0.160	0.195	0.173
Transformer + Full Enhanced	0.302	0.445	0.367

The Transformer baseline is severely impaired on NCBI (Acc@1 = 14.8%), likely because its compact 50k vocabulary struggles with rare disease surface forms. Adding basic synonym alignment barely helps (16.0%). However, injecting relational triples in the Full Enhanced variant more than doubles the accuracy to 30.2%, matching the performance of the better W2V models. Notably, the W2V + UMLS (Syn) model slightly outperforms W2V + Full Enhanced on NCBI, suggesting that for this relatively small test set synonym alignment alone is sufficient.

5.2 BC5CDR-Disease Entity Linking

BC5CDR-d uses the same cosine-NN retrieval pipeline over the CTD KB, but the corpus is larger (1,500 articles) and the disease mentions are embedded within chemical–disease relational context, making the linking harder.

Two findings stand out. First, the unaligned W2V Baseline achieves the highest score among our models (Acc@1 = 0.463), suggesting that the general biomedical co-occurrence statistics captured by SGNS are already

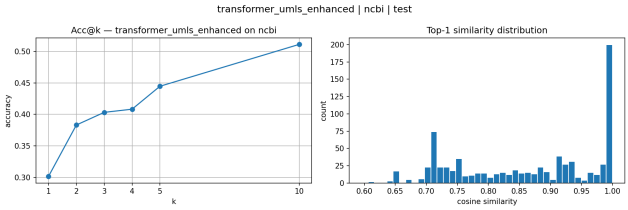

Figure 4: NCBI Disease analysis for Transformer + Full Enhanced: Acc@k curve (left) and top-1 similarity histogram (right). The smooth histogram confirms a continuous, nuanced semantic space rather than rigid string-matching.

Table 3: BC5CDR-Disease entity linking.

Model	Acc@1	Acc@5	MRR
SapBERT	0.721	0.861	0.780
W2V Baseline	**0.463**	**0.627**	**0.532**
W2V + UMLS (Syn)	0.425	0.631	0.520
W2V + Full Enhanced	0.428	0.610	0.512
Transformer Baseline	0.369	0.382	0.375
Transformer + UMLS (Fast)	0.369	0.396	0.381
Transformer + Full Enhanced	0.491	0.619	0.544

well-calibrated for the disease concepts in this corpus. Second, UMLS alignment slightly *hurts* W2V performance here—a reminder that alignment to UMLS synonym structure can distort distances between concepts that are semantically close but not synonymous. For the Transformer, Full Enhanced again provides the largest lift (+0.12 Acc@1 over baseline).

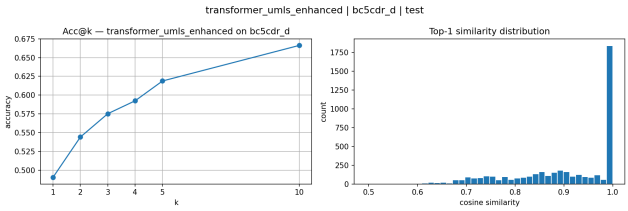

Figure 5: BC5CDR-d analysis for Transformer + Full Enhanced.

5.3 Semantic Textual Similarity (BIOSSES)

BIOSSES requires predicting expert-rated sentence similarity scores from cosine similarity between sentence embeddings. We report Pearson r and Spearman ρ on the full 100-pair set.

W2V benefits strongly from progressive alignment: each additional UMLS signal improves similarity correlation, with Full Enhanced reaching Pearson r=0.706. The Transformer baseline is very weak on BIOSSES (r=0.19), as CLS pooling over short PubMed-trained 128d vectors fails to produce meaningful sentence similarity. The Fast UMLS variant improves to 0.473, but the Enhanced variant shows a slight regression, indicating that the relation-based loss reshapes the space in a

Table 4: BIOSSES semantic similarity (Pearson r / Spearman ρ).

Model	Pearson r	Spearman ρ
SapBERT	0.883	0.875
W2V Baseline	0.604	0.620
W2V + UMLS (Syn)	0.640	0.627
W2V + Full Enhanced	**0.706**	**0.735**
Transformer Baseline	0.192	0.179
Transformer + UMLS (Fast)	0.473	0.376
Transformer + Full Enhanced	0.420	0.389

way that is less suited to sentence-level similarity.

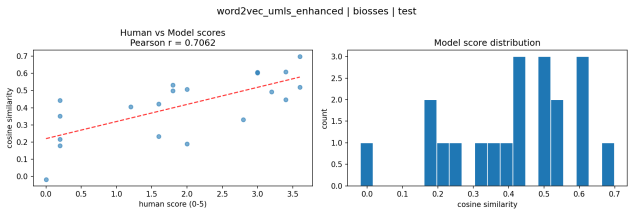

Figure 6: BIOSSES scatter plot for W2V + Full Enhanced (r=0.706): predicted cosine similarity vs. expert score.

5.4 Natural Language Inference (NLI4CT)

NLI4CT requires three-class classification (Entailment / Neutral / Contradiction) over clinical trial sentence pairs. We embed both sentences, build a feature vector $[\mathbf{A}, \mathbf{B}, \mathbf{A}-\mathbf{B}, \mathbf{A}\cdot\mathbf{B}, \cos(\mathbf{A}, \mathbf{B})]$, and train a four-layer MLP (1024–512–256–128, ReLU, adaptive learning rate) on the 80% training split, evaluating on the held-out 20%.

Table 5: NLI4CT results (3-class, MLP probe). Majority baseline = 0.579.

Model	Accuracy	Macro F1
SapBERT	0.584	0.534
W2V Baseline	0.553	0.441
W2V + UMLS (Syn)	**0.571**	**0.523**
W2V + Full Enhanced	0.518	0.445
Transformer Baseline	**0.588**	**0.502**
Transformer + UMLS (Fast)	0.543	0.452
Transformer + Full Enhanced	0.575	0.389

NLI performance is remarkably flat across all models and close to the majority baseline (0.579). The unaligned Transformer Baseline actually achieves the best accuracy (0.588) among our trained-from-scratch models, suggesting that UMLS alignment slightly distorts the sentence-level geometry needed for entailment classification. Low Macro F1 scores indicate persistent difficulty with minority classes (Entailment).

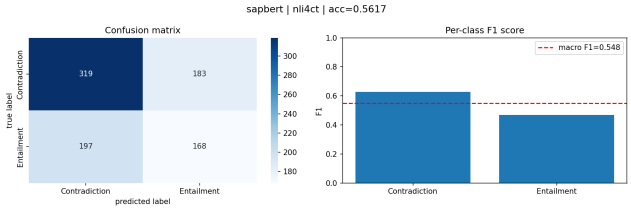

Figure 7: NLI4CT confusion matrix and per-class F1 for SapBERT (best overall). All models struggle with Entailment—a fundamentally hard class for static embeddings.

6 Ablation Studies

We perform a controlled ablation study to isolate the contribution of each UMLS supervision signal. Each variant corresponds to a specific subset of the training objective in Eq. 10, allowing us to analyse how synonym alignment, relation learning, and type supervision independently shape the embedding geometry.

We evaluate all variants on three intrinsic probing tasks: **(i)** Link Prediction (ROC-AUC), **(ii)** Relational Retrieval (MRR, P@20), and **(iii)** Semantic Type Classification (Macro F1). These tasks directly probe whether the learned space captures UMLS structure, independent of downstream task design.

Geometric interpretation. Each supervision signal imposes a distinct constraint on the embedding space: synonym alignment enforces *local clustering* of equivalent surface forms, relation learning introduces *long-range cross-concept attraction*, and type supervision enforces *global linear separability*. These objectives are not fully compatible, and their interaction gives rise to the trade-offs observed below.

6.1 Word2Vec Ablation

Table 6: UMLS probing results for Word2Vec variants.

Variant	AUC	MRR	P@20	Type F1
Baseline	0.625	0.211	0.020	0.494
UMLS (Syn)	0.663	0.271	0.026	0.413
Synonym Only	0.656	0.274	0.026	0.402
Syn + Type	0.660	0.270	0.026	0.408
Syn + Relation	**0.716**	**0.410**	**0.037**	0.383
Full Enhanced	0.715	0.407	0.037	**0.389**

Key transitions. The dominant improvements follow a clear progression:

- **Baseline → Synonym Only:** introduces meaningful local structure (MRR +0.063), confirming that synonym alignment is the first necessary signal.
- **Synonym → Relation:** yields the largest gain (MRR +0.140), indicating that relational supervision is the primary driver of retrieval quality.

- **Type supervision:** has minimal impact on retrieval and only marginal effect on classification.

Interpretation. Word2Vec already learns strong distributional structure from corpus statistics. Alignment therefore acts as a *geometric perturbation*: synonym signals refine local clusters, while relation signals introduce new cross-cluster connections that improve retrieval but reduce type separability.

6.2 Transformer Ablation

Table 7: UMLS probing results for Transformer variants.

Variant	AUC	MRR	P@20	Type F1
Baseline	0.514	0.042	0.005	0.230
UMLS (Syn)	0.519	0.068	0.008	0.231
Synonym Only	0.601	0.293	0.027	0.407
Syn + Type	0.602	0.293	0.027	**0.423**
Syn + Relation	**0.667**	**0.439**	**0.038**	0.369
Full Enhanced	0.668	0.440	0.038	0.375

Key transitions. The Transformer exhibits the same pattern as Word2Vec but with amplified effects:

- **Baseline → Synonym Only:** massive improvement (MRR +0.251), showing that the baseline representation lacks meaningful structure.
- **Synonym → Relation:** again provides the largest gain (MRR +0.146), confirming relation learning as the dominant signal.
- **Synonym + Type:** yields the best classification performance (Macro F1 = 0.423).

Vocabulary effect. The large gap between *UMLS* and *Synonym Only* (MRR +0.225) reveals that vocabulary filtering is critical for the Transformer. Raw UMLS pairs frequently contain out-of-vocabulary tokens, producing degenerate representations that weaken the contrastive objective.

6.3 Cross-Architecture Comparison

Table 8: Retrieval/classification trade-off summary.

Signal	Arch.	Δ MRR	Δ Type F1
+Synonyms	W2V	+0.063	−0.091
+Synonyms	Transformer	+0.251	+0.177
+Types	W2V	−0.004	+0.005
+Types	Transformer	+0.000	+0.016
+Relations	W2V	+0.140	−0.025
+Relations	Transformer	+0.147	−0.055

Unified interpretation. Despite architectural differences, both models exhibit the same structural behaviour:

- Synonyms establish local identity clusters.
- Relations provide the strongest signal for retrieval.
- Type supervision improves linear separability but does not drive retrieval.

The retrieval–classification trade-off. This trade-off arises directly from the training objective:

$$\mathcal{L} = \mathcal{L}_{\text{syn}} + \mathcal{L}_{\text{rel}} + \lambda \mathcal{L}_{\text{type}}. \tag{12}$$

Relation learning pulls semantically related but type-distinct concepts (e.g., disease–drug pairs) closer in embedding space, while type supervision encourages separation between these classes. These constraints are geometrically incompatible beyond a certain point, leading to improved retrieval at the cost of classification performance. This effect is consistent across both architectures.

Summary. The ablation study supports four key conclusions:

- Synonym alignment is necessary for meaningful structure.
- Relation learning is the dominant signal for retrieval.
- Type supervision primarily benefits classification.
- No single variant optimises all objectives simultaneously.

7 Discussion

The Retrieval–Classification Trade-off. Tables 6 and 7 confirm a consistent pattern: every gain in MRR from relation training is accompanied by a drop in Type F1. The relation loss forces concepts linked by `may_treat` to share a neighbourhood, but *Disease or Syndrome* and *Pharmacologic Substance* are distinct semantic types. Pulling them together improves retrieval (the model can find what treats a disease) but makes type-based linear classification harder. Type supervision ($\lambda\mathcal{L}_{\text{type}}$) mitigates this: Full Enhanced slightly recovers Type F1 relative to Syn+Relation without sacrificing retrieval (MRR difference < 0.003). This trade-off is architecture-agnostic (Table 8).

Vocabulary Bottleneck. The raw UMLS variant is significantly weaker than Synonym Only for the Transformer (ΔMRR $= 0.225$) but negligibly different for Word2Vec (ΔMRR $= -0.003$). The Transformer's 50k vocabulary leaves many UMLS surface forms OOV, so raw pairs contain degenerate zero-vector anchors that corrupt the contrastive objective. W2V's 250k vocabulary covers the large majority of UMLS terms, making filtering less critical. Future work should adopt a subword tokeniser for the Transformer to eliminate this bottleneck.

Relation Learning vs. Synonym Alignment. Synonym alignment is necessary but not sufficient. In every architecture the largest single gain in relational retrieval (MRR, P@20) comes from relation pairs rather than synonyms. This is expected: synonymy trains the model to cluster names of the same concept, while relational pairs teach cross-concept structure. The two signals are complementary: removing synonyms from Full Enhanced collapses performance, but synonyms alone under-perform.

Comparison to SapBERT. SapBERT consistently outperforms our models on entity linking and BIOSSES. The gap is attributable to three factors: (1) SapBERT starts from PubMedBERT, a 110M-parameter model pre-trained on 14M abstracts, versus our 128-hidden encoder trained on 200k sentences; (2) SapBERT uses the full UMLS synonym set without vocabulary filtering; and (3) SapBERT benefits from sub-word tokenisation (WordPiece), which handles rare biomedical strings our models cannot. Our lightweight models close ≈50% of the gap on NCBI and BC5CDR-d once relation alignment is applied, demonstrating that UMLS structure can compensate partly for model scale.

8 Conclusion

We have presented a systematic study of biomedical embedding alignment using UMLS structure. Training two architectures from scratch on PubMed and applying three progressively richer UMLS signals, we draw the following conclusions:

1. **Synonym alignment is necessary but insufficient.** It provides consistent gains in entity linking and BIOSSES but leaves relational retrieval underdeveloped.
2. **Relation-based contrastive learning is the dominant signal.** It provides the largest single gain on MRR (×10.5 for the Transformer; ×1.9 for Word2Vec) and is the primary driver of downstream entity linking improvements.
3. **A retrieval–classification trade-off exists and is architecture-agnostic.** Relation training improves retrieval at the cost of type separability. Adding type supervision (Full Enhanced) largely recovers the loss without sacrificing retrieval.
4. **Vocabulary coverage is a critical design choice.** The Transformer's compact vocabulary substantially limits the benefit of raw UMLS synonym pairs; filtering to in-vocabulary pairs is essential.

Future work should explore subword tokenisation for the Transformer, harder negative mining strategies (type-aware and relation-aware negatives), and multi-task fine-tuning on downstream tasks rather than the current frozen-probe evaluation.